

## **Job Title: Junior Backend Developer**

**Department: Earth Observation Data Department (EODD)**

**Egyptian Space Agency (EgSA)**

### **1. Job Description**

The Egyptian Space Agency (EgSA) invites qualified and motivated candidates to join the Earth Observation Data Department (EODD). The successful candidate will contribute to the development, maintenance, and enhancement of backend software systems that support Earth Observation applications, satellite data processing, and geospatial services.

Working under the guidance of senior developers, the selected candidate will participate in developing backend services, processing workflows, APIs, and databases while gaining hands-on experience with modern software engineering practices and Earth Observation technologies.

### **2. Responsibilities**

- Develop and maintain backend components for Earth Observation applications.
- Assist in developing data processing workflows for satellite and geospatial data.
- Develop and maintain RESTful APIs to support web and desktop applications.
- Participate in integrating multiple data sources into backend systems.
- Support database design, implementation, and maintenance.
- Write clean, maintainable, and well-documented code following software development best practices.
- Participate in testing, debugging, and troubleshooting software applications.
- Assist in improving system performance, reliability, and scalability.
- Collaborate with software developers, GIS specialists, and Earth Observation experts to deliver integrated solutions.
- Participate in code reviews and continuously improve coding skills.
- Stay up to date with emerging backend technologies and software development practices.

### **3. Requirements**

#### **3.1. Education**

- Bachelor's degree in Computer Science, Computer Engineering, Software Engineering, Information Systems, Artificial Intelligence, Data Science, Remote Sensing, Geomatics, Geography, or a related field..

#### **3.2. Technical Skills**

- Good programming skills in **Python**.
- Knowledge of object-oriented programming (OOP) concepts.
- Basic understanding of relational databases such as PostgreSQL, MySQL, or SQL Server.
- Basic knowledge of REST APIs and web services.
- Familiarity with version control systems (Git).
- Basic understanding of Linux operating systems is an advantage.
- Familiarity with backend frameworks such as **FastAPI**, **Flask**, or **Django** is preferred.
- Basic knowledge of cloud platforms or Docker is considered an advantage.

- Familiarity with Earth Observation or GIS technologies is a plus but not required.

### **3.3. Personal Skills**

- Strong analytical and problem-solving skills.
- Willingness to learn new technologies and programming tools.
- Ability to work effectively within a multidisciplinary team.
- Good written and verbal communication skills.
- Good command of English.
- Ability to manage assigned tasks and meet deadlines.

## **4. Preferred Qualifications**

Candidates with experience in one or more of the following areas will have an advantage:

- Geospatial data processing.
- Python libraries such as **GeoPandas**, **Rasterio**, or **GDAL**.
- PostgreSQL/PostGIS.
- Docker.
- Agile software development practices.
- Earth Observation or Remote Sensing applications.